Supplementary Material for

Adaptive Temporal Action Quantization for Vision-Language-Action Models

A Additional Experiments

A.1 Experiment on dexterous hand embodiment

The main motivation of ATQ is that teleoperated demonstrations inherit conservative execution patterns from the human operator, and that this inefficiency enables adaptive temporal quantization. This effect is expected to be even more pronounced on high-DoF dexterous hand embodiments, through a teleoperation interface forces the operator to move more precisely and slowly than gripper based robots. To examine whether ATQ translates these slower demonstrations into larger execution speedups while preserving task success, we evaluate GR00T N1.5 [1] and GR00T N1.5 + ATQ on an OpenArm robot equipped with a 6-DoF Inspire RH56F1 dexterous hand. See Figure 5b for the hardware setup, Appendix C.2 for detailed descriptions, and implementation details in Appendix C.3:

- **Pick Bottle to the Shelf.** A robot must pick up a target bottle from the table and place it on the shelf. To assess generalization and language following, we use a diverse set of bottle instances and four target placement positions on the shelf.

Table 1 shows that ATQ reduces the number of executed action steps and improves the success rate over GR00T N1.5, while fixed-ratio quantization shortens execution at a substantial cost in success rate. This is consistent with our hypothesis that for the same distance, the base policy must generate many action chunks, and each chunk introduces another chance of an action error. ATQ covers the same distance with fewer chunks, lowering the cumulative action error along a trajectory and raising success. Fixed quantization applies the same quantization ratio to every chunk and cannot decide where to quantize more or less, so its success rate substantially degrades. These results suggest that ATQ transfers across embodiments and scales with the temporal redundancy in the underlying demonstrations.

Table 1: **Real-World dexterous hand evaluation results.** We compare ATQ with baseline methods on the *Pick Bottle to the Shelf* task, performed on an OpenArm robot equipped with a 6-DoF Inspire RH56F1 dexterous hand. We report the average success rate (%, over 24 trials) and Succ-steps (average action-steps on successful trials, lower is better).

Method	*Pick Bottle to the Shelf*	
	Avg. ↑	Succ-steps ↓
GR00T N1.5	35.4	483.3
+ fixed quantization Q=2	33.3	371.4
+ fixed quantization Q=3	20.8	384.6
+ ATQ (Ours)	**54.2**	397.6

B Additional Analysis

B.1 Quantitative analysis on Ground-Truth Data

Setup. We ask whether training dedicated coarser decoders is worthwhile, or whether one could obtain equally good coarse actions by simply quantizing a standard policy's fine-grained actions. We test this on RoboCasa-Kitchen simulation benchmark [2]. We sample trajectory points across all 24 tasks. We feed the ground-truth observation to each policy: our ATQ policy, and the GR00T N1.5 [1] baseline, both fine-tuned on the same data. We made the comparisons only at the points where our confidence-gated router (τ_c=0.7) selects a coarser decoder. Method for producing the quantized action chunk:

- **Ours**: the prediction of the coarser decoder D_k that the router selected.

Table 2: Accuracy of quantized action chunks against the ground-truth quantized target on RoboCasa, on points where the confidence-gated router(τ_c=0.7) selected a coarser (quantizing) decoder (n=1,234). *Ours* is our trained coarser decoder; *Base* is the GR00T N1.5 baseline's 16-step output quantizeded with the same rule. Each coarser decoder spec is represented as (Action horizon, Quantization ratio)

Decoder	Metric	Ours	Base
(16,2)	MSE ↓	0.0209	0.0240
	DTW ↓	3.087	3.383
	cosine ↑	0.896	0.879
(8,2)	MSE ↓	0.0200	0.0258
	DTW ↓	1.512	1.766
	cosine ↑	0.895	0.871
Both	MSE ↓	0.0204	0.0249
	DTW ↓	2.289	2.564
	cosine ↑	0.896	0.875

- **Base**: the GR00T N1.5 baseline's raw 16-step prediction, then naively block-aggregated with the same rule used to build D_k's target (delta dimensions block-summed, latching dimensions block-last).

Both are compared against the ground-truth quantized target $\mathbf{z}^{(k)}$. Metrics are computed on the continuous action dimensions only, since the latching gripper/mode dimensions are categorical.

Metrics. Let $\hat{\mathbf{z}} = (\hat{\mathbf{z}}_1, \ldots, \hat{\mathbf{z}}_T)$ be a predicted quantized chunk and $\mathbf{z} = (\mathbf{z}_1, \ldots, \mathbf{z}_T)$ the GT quantized target, with $\mathbf{z}_i \in \mathbb{R}^d$ over the continuous dimensions. We report three complementary measures. The mean squared error captures pointwise magnitude error;

$$\mathrm{MSE}(\hat{\mathbf{z}}, \mathbf{z}) = \frac{1}{Td} \sum_{i=1}^{T} \|\hat{\mathbf{z}}_i - \mathbf{z}_i\|_2^2, \tag{1}$$

The cosine similarity on the flattened chunks, captures directional agreement;

$$\cos(\hat{\mathbf{z}}, \mathbf{z}) = \frac{\mathrm{vec}(\hat{\mathbf{z}})^\top \mathrm{vec}(\mathbf{z})}{\|\mathrm{vec}(\hat{\mathbf{z}})\| \, \|\mathrm{vec}(\mathbf{z})\|}, \tag{2}$$

The dynamic time warping (DTW) captures shape agreement under small temporal misalignment by aligning the predicted and target latent sequences,

$$\mathrm{DTW}(\hat{\mathbf{z}}, \mathbf{z}) = D(T,T), \quad D(i,j) = \|\hat{\mathbf{z}}_i - \mathbf{z}_j\|_2 + \min\big\{D(i\text{–}1, j),\ D(i, j\text{–}1),\ D(i\text{–}1, j\text{–}1)\big\}, \tag{3}$$

with $D(0,0) = 0$ and $D(i,0) = D(0,j) = \infty$.
Lower is better for MSE and DTW, higher for cosine similarity.

Results. Table 2 reports averages on each metrics of our decoders. At the points where the router selected a coarser decoder, our trained decoders match the GT quantized target more closely than naively quantizing the base model on all three metrics. The advantage holds for both coarser decoders.

Interpretation. These results indicate that the coarser decoders are not redundant with post-hoc quantization: a decoder trained to predict the quantized action directly reproduces the ground-truth quantized motion more faithfully than quantizing a baseline policy's fine-grained output. This isolates the contribution of the learned decoders and supports training dedicated decoders for each horizon rather than quantizing a single shared output.

B.2 Qualitative analysis on Ground-Truth Trajectory

B.2.1 Ground-Truth Action Trajectory with Quantization

Figures 1 and 2 the cumulative end-effector trajectory of the original action chunk against its 2× quantized counterpart, for the real-robot and Robocasa-Kitchen simulation datasets respectively. Each

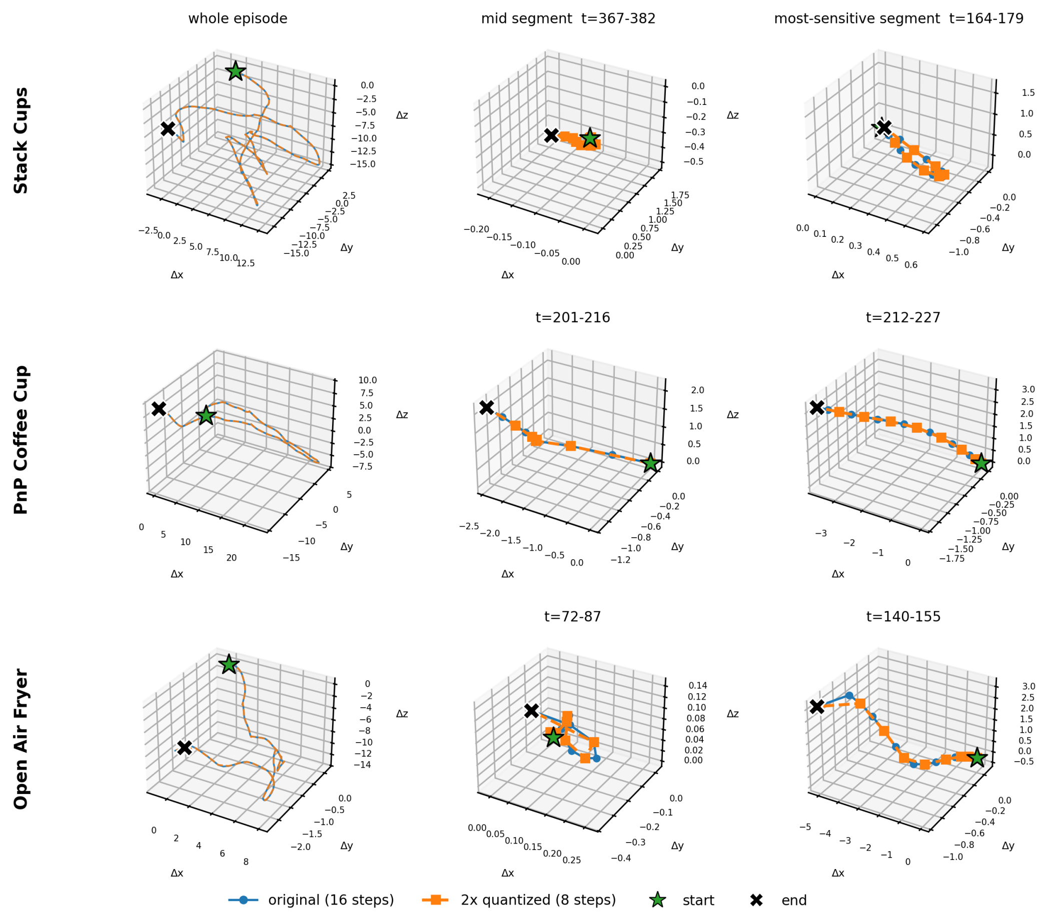


Figure 1: **Real-robot trajectories under** $2\times$ **quantization.** For each task (row), we show the cumulative end-effector trajectory of the original action chunk (blue) and its $2\times$ quantized counterpart (orange) at three zoom levels: the whole episode (left), a representative 16-step chunk from the middle of the episode (center), and the chunk with the largest per-chunk approximation error (right). The quantized trajectory closely follows the original even in the most-sensitive segment.

row is one task and shows three views at increasing zoom. The left panel plots the *whole episode*: because $2\times$ quantization preserves the endpoint of every quantized block, the quantized trajectory (orange) tracks the original (blue) almost exactly at this scale, confirming that the net motion is retained. Since a full episode therefore reveals little, the middle panel zooms into a single 16-step chunk taken from the middle of the episode. The right panel shows the chunk within the episode where the quantized trajectory departs most from the original, i.e. the worst case for quantization in that trajectory. On the real-robot tasks (Figure 1) the quantized trajectory stays close to the original even in this worst-case segment, departing from it only slightly; The simulation trajectories (Figure 2) are somewhat less smooth and deviate a little more, though they remain close overall. This implies that applying chunk-based quantization without hindering the overall trajectory progress may not have a significant impact on task success.

B.2.2 Rollout Trajectory with Adaptive Temporal Quantization

Figure 3 overlays the end-effector trajectories of the GR00T-N1.5 baseline and GR00T-N1.5 + ATQ on four RoboCasa tasks, rolled out from the same seed so each pair starts from an identical state. The right panel of each task zooms into the point where the two paths first separate by a small fixed distance in the simulator's world-frame metric. ATQ reaches the goal in markedly fewer environment steps while following a path of broadly similar shape, illustrating that its quantized, multi-step actions

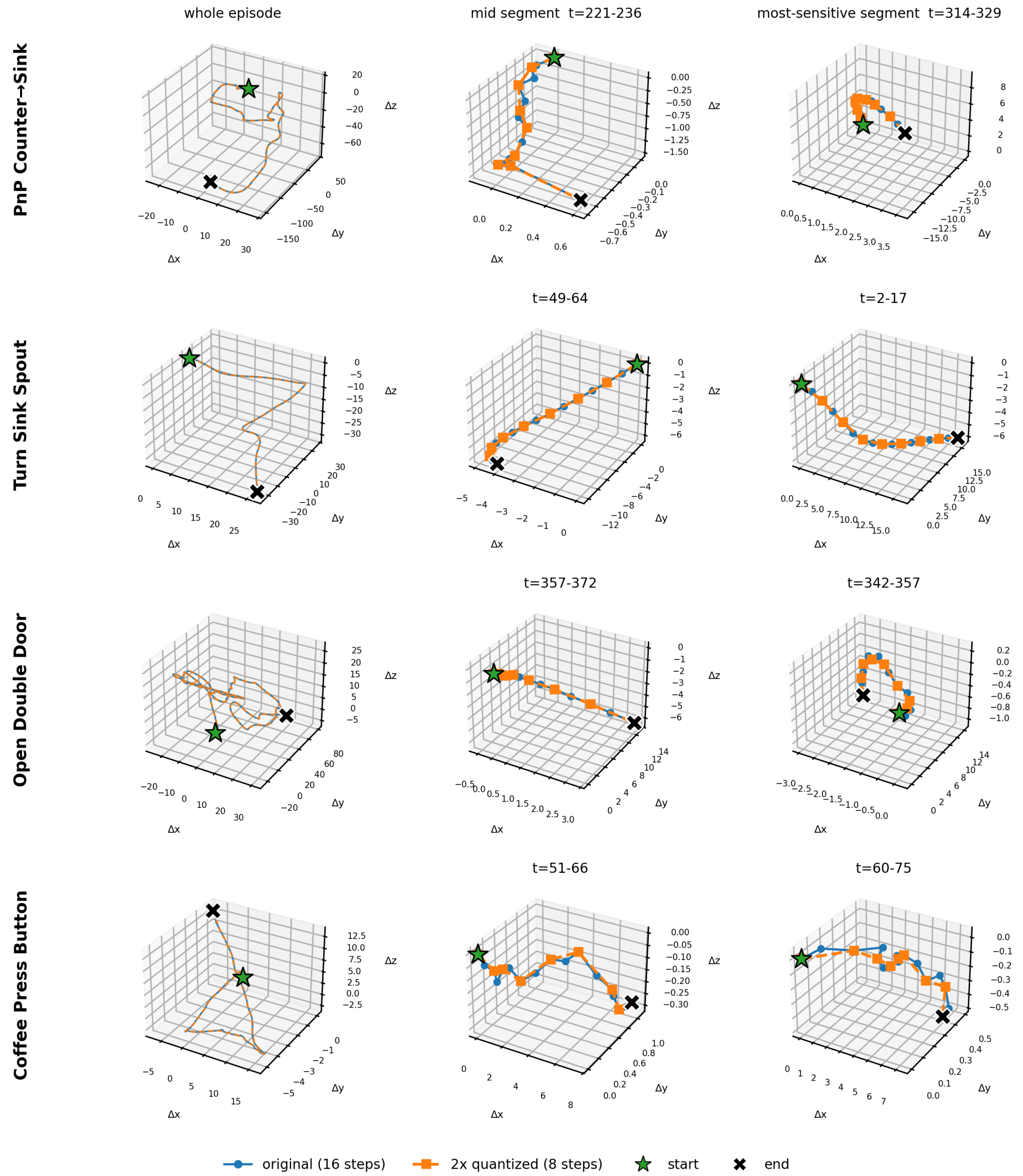


Figure 2: **RoboCasa trajectories under** $2\times$ **quantization.** Same visualization as Figure 1 for the simulation dataset. The simulation actions are somewhat less smooth, so the quantized trajectory deviates slightly more from the original in the most-sensitive segments, though it remains close overall.

remain consistent with task completion. We show this only as a qualitative illustration: closed-loop rollouts are stochastic, so even re-running the same baseline may not reproduce an identical trajectory.

B.3 Further Qualitative Analysis

We further show the probability of selecting the action decoder that produces quantized actions over time on additional tasks including Real-World task. Across these tasks we observe a broadly consistent pattern: the probability stays in its upper range during free-space transit and tends to drop around phases that are intuitively harder to quantize, namely those requiring precise positioning or contact.

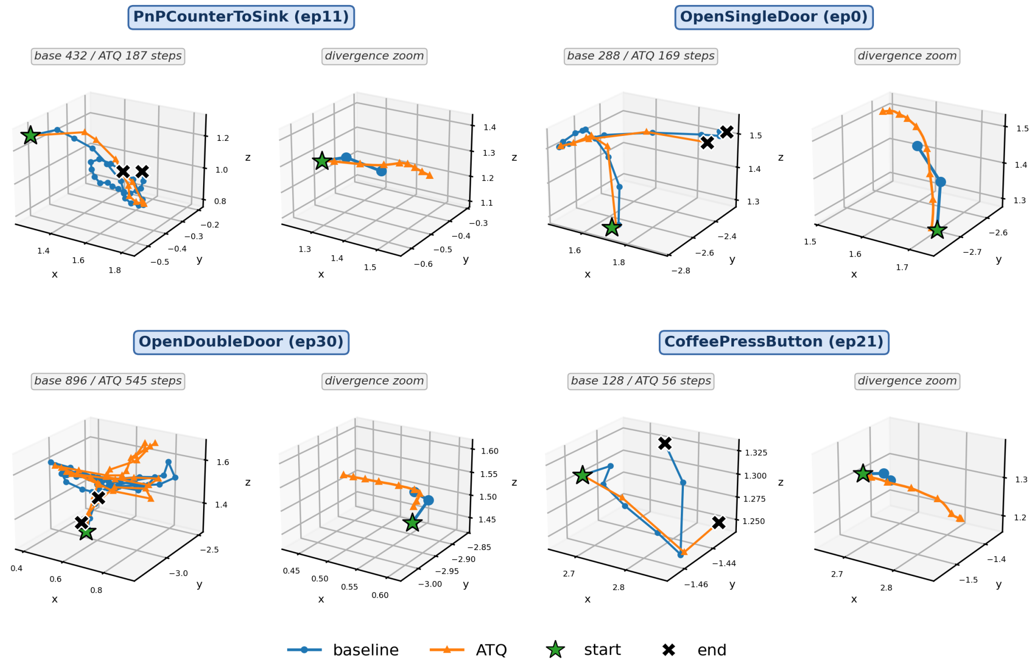


Figure 3: **End-effector rollout trajectories of the GR00T-N1.5 baseline and GR00T-N1.5 + ATQ.** For four representative RoboCasa tasks, we overlay the absolute end-effector position trajectories produced. Within each task, the *left* panel shows the whole-episode trajectory and the *right* panel (*divergence zoom*) is a close-up of the $\sim$32-step window where the two trajectories first departs from the baseline.

C Experimental Details

C.1 Hardware Details

We visualize the hardware setup of our real-robot platforms in Figure 5. For the main experiments, we construct a Franka Research 3 platform, a 7-DoF single arm robot equipped with a Robotiq 2F-85 gripper, following the DROID [3] setup. To further demonstrate the applicability of our method across different embodiments, we additionally consider the OpenArm robot equipped with an Inspire RH56F1 Hands, a 6-DoF dexterous hand.

C.2 Real-Robot Experiment Details

For training, we collected 60 datasets per tasks for Gripper embodiment and 100 datasets for dexterous hand embodiment.

- **Gripper Stack 3 Cups**: We evaluate each model over 24 trials. We use the instruction *"Stack the cups."* for this task.
- **Gripper PnP Coffe Cup**: We evaluate each model over 24 trials, consisting of 6 trials with 2 kinds of cups (blue and brown cups) and 2 locations (left and right) of coaster. We use the instructions as *"pick (blue/brown) cup from machine and place to (left/right) coaster."* for this task. We evaluated the rollout as a failure if the cup went more than halfway off the coaster.
- **Gripper Open AirFryer**: We evaluate each model over 24 trials, with 2 different location of AirFryer. We use the instruction *"Open the AirFryer."* for this task.
- **Dexterous Hand Pick Bottle to the Shelf**: we evaluate each model over 4 different target object categories with 6 trials per category, resulting in 24 evaluation trials in total. We made 4 shelf

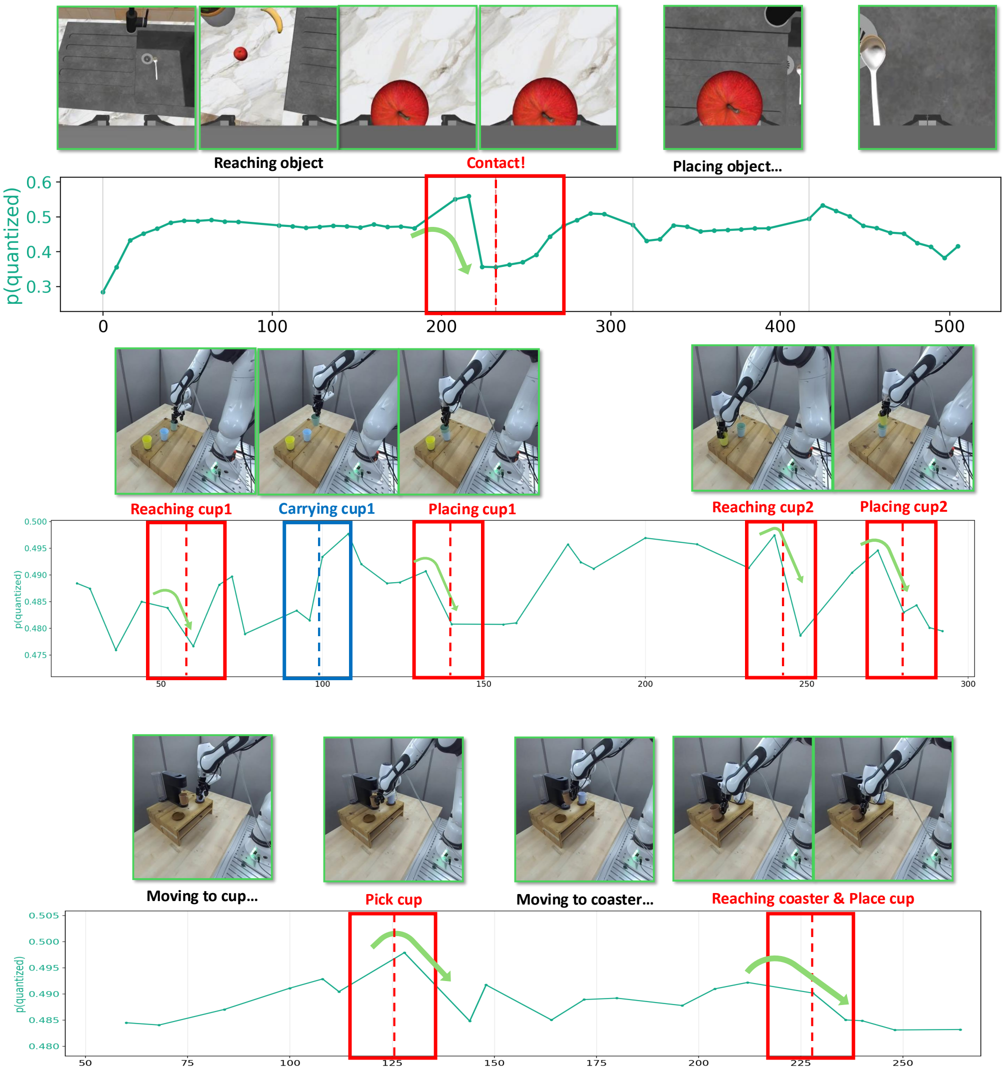
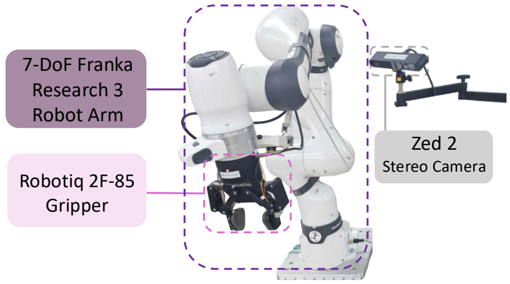


Figure 4: **Further Router pick analysis.** We show the probability of selecting the action decoder that produces quantized actions over time, shown alongside the corresponding scenes from Robocasa-Kitchen simulation and real world tasks.

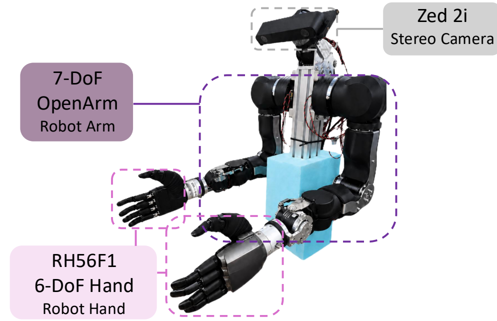
locations (top-right, top-left, bottom-right, bottom-left), and allocated different locations to each categories. We use the instructions as *”Move the (tea/trevi/water/pokari sweat) bottle to (top right, top left, bottom right, bottom left).”* for this task.

C.3 Implementation Details

We report the performance of ATQ by fine-tuning the pretrained Vision-Language-Action model GR00T N1.5, following its official implementation. We freeze the VLM backbone and only train the action expert. All models are trained jointly on a consolidated dataset combining the training data from all tasks, with a batch size of 64 for 60k iterations. We set the coefficient of the balance loss to $\lambda_{\text{bal}} = 0.05$ and KL loss to $\lambda_{\text{sup}} = 0.1$. We set the threshold to κ=0.7. The remaining training hyperparameters are summarized in Table 3.



(a) **Franka Research 3 gripper platform.** A 7-DoF Franka Research 3 robot arm equipped with a Robotiq 2F-85 gripper, together with a wrist-mounted Zed Mini stereo camera for visual observation.



(b) **Bimanual dexterous-hand platform.** Dual 7-DoF OpenArm robot arms, each equipped with a 6-DoF RH56F1 dexterous hand, together with a head-mounted Zed 2i stereo camera for visual observation.

Figure 5: **Overview of our hardware platforms.** We use two robot setups

Table 3: Hyperparameter details for training ATQ.

	GR00T N1.5 + ATQ
Optimizer	AdamW
Optimizer momentum	$\beta_1, \beta_2 = 0.95, 0.999$
Optimizer weight decay	1e-5
Learning rate	1e-4
Learning rate scheduler	Cosine decay
Warmup iterations	3,000
Batch size	64
Training iterations	60,000

D More Quantitative Results

In this section, we report detailed task-wise results for the main experiments. Table 4 reports per-task success rates on RoboCasa-Kitchen simulation for GR00T N1.5 trained with and without ATQ, while Table 5 reports per-task success rates DexJoco under the same comparison.

Table 4: Per-task success rates (%) on RoboCasa-Kitchen (24 tasks). We compare GR00T N1.5 against GR00T N1.5 + ATQ at 30, 100, and 300 demonstrations.

Task	GR00T N1.5			GR00T N1.5 + ATQ		
	30	100	300	30	100	300
RoboCasa Kitchen (24 tasks, PnP = Pick-and-Place)						
Close Double Door	34.0	80.0	78.0	40.0	80.0	80.0
Close Drawer	94.0	94.0	96.0	94.0	96.0	96.0
Close Single Door	92.0	100.0	94.0	96.0	96.0	98.0
Coffee Press Button	78.0	72.0	94.0	72.0	78.0	86.0
Coffee Serve Mug	56.0	72.0	60.0	72.0	64.0	70.0
Coffee Setup Mug	22.0	22.0	30.0	20.0	38.0	50.0
Open Double Door	64.0	80.0	84.0	74.0	76.0	82.0
Open Drawer	42.0	62.0	70.0	42.0	58.0	64.0
Open Single Door	58.0	60.0	74.0	54.0	68.0	70.0
PnP Cab → Counter	30.0	56.0	60.0	28.0	54.0	62.0
PnP Counter → Cab	30.0	54.0	54.0	38.0	56.0	64.0
PnP Counter → Microwave	26.0	50.0	46.0	30.0	44.0	40.0
PnP Counter → Sink	34.0	62.0	72.0	28.0	60.0	74.0
PnP Counter → Stove	44.0	64.0	70.0	34.0	68.0	76.0
PnP Microwave → Counter	32.0	50.0	34.0	30.0	56.0	52.0
PnP Sink → Counter	48.0	56.0	56.0	50.0	54.0	56.0
PnP Stove → Counter	38.0	58.0	58.0	36.0	62.0	68.0
Turn Off Microwave	64.0	88.0	88.0	78.0	88.0	92.0
Turn Off Sink Faucet	66.0	84.0	94.0	52.0	74.0	94.0
Turn Off Stove	12.0	6.0	8.0	12.0	24.0	22.0
Turn On Microwave	42.0	48.0	52.0	48.0	68.0	62.0
Turn On Sink Faucet	72.0	84.0	80.0	76.0	86.0	88.0
Turn On Stove	26.0	54.0	44.0	26.0	52.0	36.0
Turn Sink Spout	42.0	64.0	74.0	42.0	72.0	68.0
Average	47.8	63.3	65.4	**48.8**	**65.5**	**68.8**

Table 5: Per-task success rate on DexJoCo single-arm (6 tasks, 50 episodes each). We compare GR00T N1.5 against GR00T N1.5 + ATQ.

Task	GR00T N1.5	GR00T N1.5 + ATQ
Hammer Nail	92.0	86.0
Click Mouse	72.0	86.0
Pick Bucket	68.0	80.0
Pinch Tongs	56.0	64.0
Fold Glasses	62.0	56.0
Water Plant	88.0	80.0
Average	73.0	**75.3**

References

[1] J. Bjorck, V. Blukis, F. Castañeda, N. Cherniadev, X. Da, R. Ding, L. Fan, Y. Fang, D. Fox, F. Hu, S. Huang, et al. Gr00t n1.5: An improved open foundation model for generalist humanoid robots. https://research.nvidia.com/labs/gear/gr00t-n1_5/, June 2025. Accessed: 2025-09-09.

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[3] A. Khazatsky, K. Pertsch, S. Nair, A. Balakrishna, S. Dasari, S. Karamcheti, S. Nasiriany, M. K. Srirama, L. Y. Chen, K. Ellis, et al. Droid: A large-scale in-the-wild robot manipulation dataset. *arXiv preprint arXiv:2403.12945*, 2024.